

ENGR 1451/2451: Exploratory Data Science

Scatter Diagrams and Correlation

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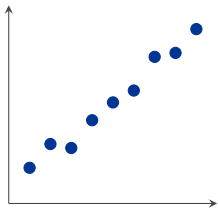
Today's Goals

Use scatter diagrams to describe relationships between two quantitative variables.

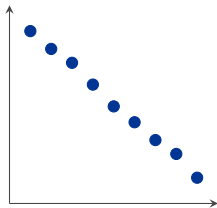
- Explain positive, negative, and weak association.
- Interpret the correlation coefficient r as a unitless measure of linear association.
- Use standard units to compare the relative positions of paired observations.
- Avoid common misinterpretations of correlation.

Direction of Association

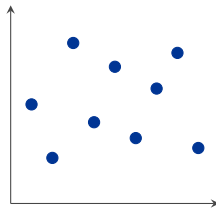
Positive



Negative



Weak / no linear

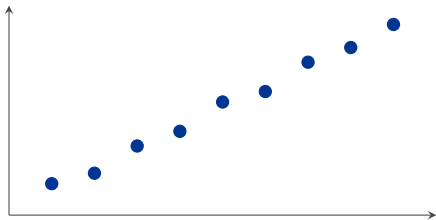


Reading direction

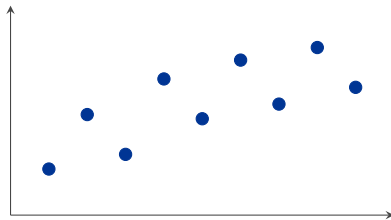
Positive means higher x tends to go with higher y . Negative means higher x tends to go with lower y .

Strength: How Tight Is the Cloud?

Strong positive association



Weak positive association



Strength is about scatter around a line

A tighter linear cloud gives a correlation closer to 1 or -1 .

Correlation Coefficient r

The **correlation coefficient** summarizes the direction and strength of a **linear** association.

Range

$$-1 \leq r \leq 1.$$

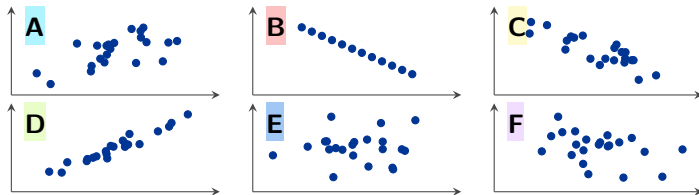
- $r > 0$: positive association
- $r < 0$: negative association
- $r \approx 0$: weak linear association

Important properties

- Unitless.
- Unchanged by switching inches to centimeters.
- Sensitive to outliers and nonlinear patterns.
- Does not prove causation.

Quick Check: Match r to the Scatterplot

Each plot has the same scale. Match each scatterplot with one correlation value. Use each value once.



Options: -1.00 , -0.85 , -0.38 , 0.06 , 0.62 , 0.97

Correlation in Standard Units

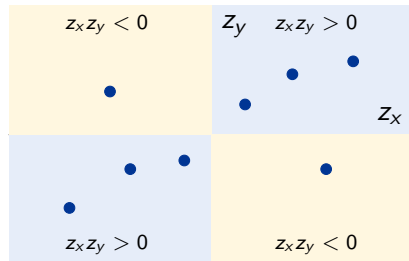
Convert each variable to standard units:

$$z_x = \frac{x - \bar{x}}{s_x}, \quad z_y = \frac{y - \bar{y}}{s_y}.$$

Sample correlation

$$r = \frac{1}{n-1} \sum_{i=1}^n z_{x,i} z_{y,i}.$$

Positive products increase r ; negative products decrease r .



Computing r in Python

Direct command

```
r = df["x"].corr(df["y"])
```

With missing values removed

```
xy = df[["x", "y"]].dropna()  
r = xy["x"].corr(xy["y"])
```

- Use a scatter plot before computing r .
- Check that both variables are quantitative.
- Missing values must be handled consistently.

Plot first

```
df.plot.scatter(  
    x="x", y="y"  
)
```

Quick Check: Sign of Correlation

Predict the sign before calculating.

1. Age of a used car and price, excluding antiques.
2. Vehicle weight and miles per gallon.
3. Hours studied and exam score.

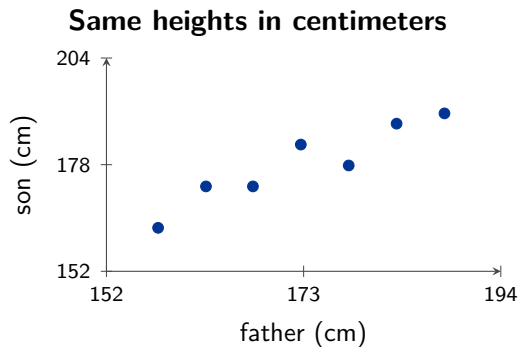
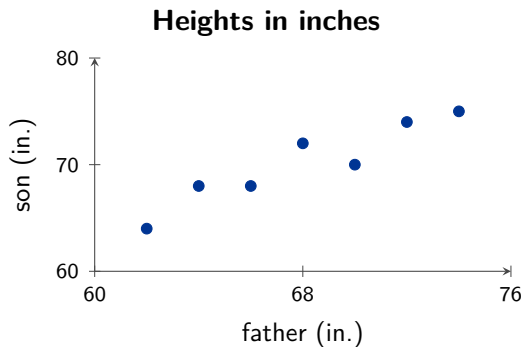
What Does $r = 0.90$ Mean?

True or false? Explain.

If the correlation coefficient is 0.90, then 90% of the points are highly correlated.

Discuss: What kind of number is r ? Is it a percentage of points?

Unit Changes Do Not Change r



Why?

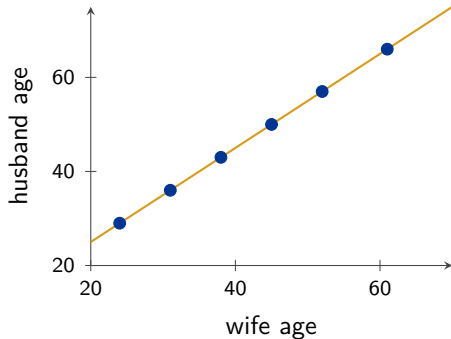
Correlation uses standard units, so shifts and positive rescalings of units cancel out.

Perfect Correlation Does Not Mean Equal Values

Suppose in a hypothetical population:

$$\text{husband's age} = \text{wife's age} + 5.$$

What is the correlation between husband age and wife age?

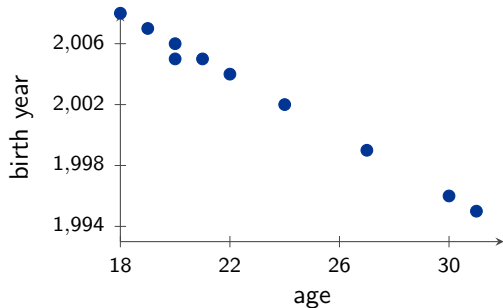


Nearly -1 Versus Exactly -1

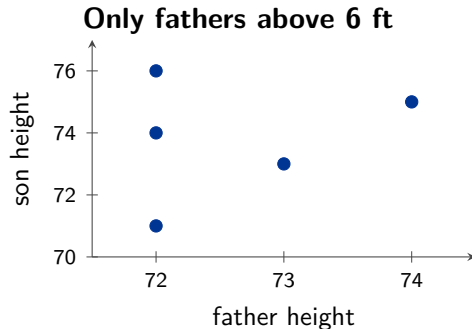
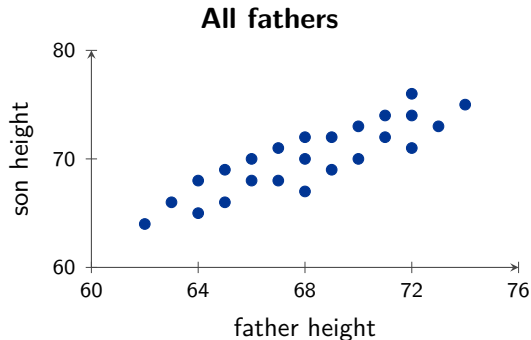
Student age and year of birth

Older students usually have earlier birth years, so the association is negative.

- If age were measured exactly at one instant, then year of birth would be almost a straight-line function.
- But birthdays, rounding to whole years, and survey timing add small deviations.
- So the correlation is usually **nearly** -1 , not necessarily exactly -1 .



Restricting the Range Can Change Correlation



Takeaway

A correlation computed on a restricted subgroup can be much smaller than the correlation in the full data.

Discussion Question: Correlation With a Sum

Investigators sampled OINKS: dual-income couples with no kids.

$$\text{family income} = \text{husband's income} + \text{wife's income}.$$

The average family income was about \$85,000. About 10% of couples had family income between \$80,000 and \$90,000.

Fill in the blanks using the choices below:

- (a) The correlation between wife's income and family income is _____.
- (b) Among couples with family income between \$80,000 and \$90,000, the correlation between wife's income and husband's income is _____.

−1 nearly −1 somewhat negative 0 somewhat positive nearly 1

Think in Standard Units

Standardize each variable separately:

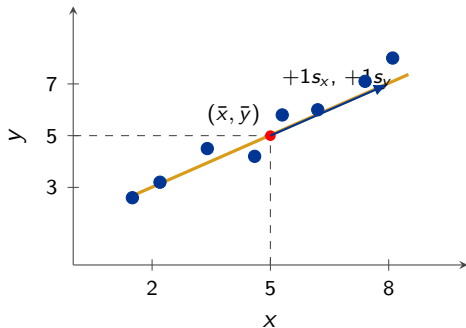
$$z_x = \frac{x - \bar{x}}{s_x}, \quad z_y = \frac{y - \bar{y}}{s_y}.$$

Equal standardized position

For a positive association, a useful reference is

$$z_y = z_x.$$

So +1 SD in x corresponds to +1 SD in y .



Standard Units Center Both Variables at Zero

True or false?

1. The point of averages $(\bar{x}, \bar{y})$ becomes $(0, 0)$ after standardizing.
2. In the original units, the point of averages must be $(0, 0)$.

Example: Match Standardized Position

In a class, two exam-score summaries are:

$$\bar{x} = 75 \text{ points}, \quad s_x = 10 \text{ points}, \quad \bar{y} = 82 \text{ points}, \quad s_y = 8 \text{ points}, \quad r = 0.60.$$

Question

A student scores 85 points on Exam 1. What Exam 2 score has the **same standardized position**?

Which Pairs Have the Same Standardized Position?

Use the same summaries:

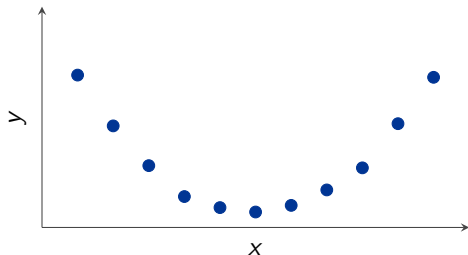
$$\bar{x} = 75, s_x = 10, \quad \bar{y} = 82, s_y = 8.$$

For each pair, compare z_x and z_y

1. Exam 1 95 points, Exam 2 98 points
2. Exam 1 65 points, Exam 2 78 points
3. Exam 1 65 points, Exam 2 74 points

Correlation Measures Linear Association

Clear curve, small r possible



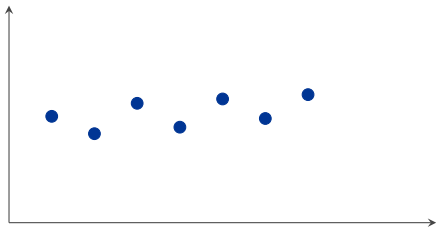
- A pattern can be strong but nonlinear.
- Correlation may miss a curved relationship.
- Always draw the scatter diagram before interpreting r .

Do not summarize blindly

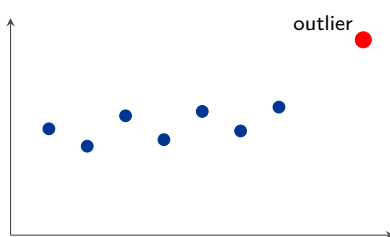
The scatter plot tells you what kind of relationship you are trying to summarize.

Outliers Can Change Correlation

Main cloud only



Same cloud plus one point



Takeaway

A single unusual point can make the linear association look stronger or weaker. Investigate outliers before relying on r .

Common Correlation Mistakes

Mistake

- Treating r as a percentage of points.
- Interpreting $r \approx 0$ as no relationship.
- Ignoring units, axes, and outliers.
- Claiming causation from correlation alone.

Better practice

- Use r as a summary of the whole cloud.
- Check for nonlinear patterns.
- Plot first, compute second.
- Use study design and subject-matter knowledge for causal claims.

Correlation is useful, but it is not a substitute for looking at the data.

Key Takeaways

- Scatter diagrams show the relationship between two quantitative variables.
- Correlation r summarizes **linear** association on a scale from -1 to 1 .
- r is unitless and is not a percentage of points.
- Restricting the range, conditioning on totals, nonlinear patterns, and outliers can change the story.
- Standard units let us compare relative position: $z_x = z_y$ means the two values are the same number of SDs from their respective averages.